An Adaptive Cooperative Caching Strategy for Vehicular Networks

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***Abstract*—Edge caching has emerged as an effective solution to the challenges posed by massive content delivery in the vehicular network. In vehicular networks, vehicles and roadside units (RSUs) can serve as intermediate relays with caching capabilities. However, due to the mobility of vehicles, the topology of the edge network changes frequently, which leads to frequent link interruptions and increases the transmission delay. This paper proposes an adaptive cooperative caching (ACC) strategy to adapt the frequent changes in the vehicular edge network topology and describes an optimization problem to minimize the average transmission delay. Then, the optimization problem is transformed into two sub-optimization problems: multiple-choice knapsack (MCK) problem and multiple minimum-weight dominating set (MMWDS) problem. Finally, two greedy algorithms with low complexity are designed to solve the above two optimization problems and obtain approximate solutions to the optimal caching decision. Simulation results show that ACC can effectively improve the cache hit rate and reduce the average transmission delay and the communication overhead compared with other caching strategies.**

***Index Terms*—Edge cooperative caching (ECC), multiple minimum -weight dominating set (MMWDS), multiple -choice knapsack (MCK), vehicular networks.**

I. Introduction

In THE past decades, intelligent vehicular networks (IVN) has been widely concerned and researched, including intelligent traffic information systems, advanced driver assistance systems, and autonomous driving [1], [2]. However, due to the high mobility of vehicles, the network topology changes rapidly. The traditional host-oriented network transmission format cannot meet the current demand for high reliability and low delay of IVN, resulting in degradation of data transmission performance. Therefore, IVN often suffers from communication interruptions [3], [4].

Due to the growing demand for communication, computing, and storage resources from end-users, lower content access delay and different application requirements may not be met if the content is obtained from remote data centers [5]. To address the above issues, edge caching is proposed to facilitate content delivery, and on-board edge computing and networking can serve as an effective framework to relieve backhaul links by migrating cloud caching servers to edge devices. In the past decade, edge caching has been proposed to serve vehicular networks, including vehicular ad-hoc network (VANET) [6], [7], [8], vehicular delay tolerant network (VDTN) [9], and especially the recently vehicular named data network (VNDN) [10], [11]. Most internet services have now gradually shifted from host-oriented to content-oriented, i.e., content requesters no longer care where the content comes from, but are more concerned with the content itself [12]. In vehicular networks, this content includes entertainment messages, traffic information, and some control and emergency messages [13]. Due to the decoupled content-location relationship in the named data network (NDN) that makes it compatible with vehicular networks with high-velocity mobility [14]. Therefore, by integrating NDN into VANET, a new content-oriented vehicular networks architecture emerged, namely, VNDN [10]. Among them, NDN has been shown to be an efficient implementation of information center network (ICN), which aims to retrieve and route the content of the message itself to accomplish fast forwarding [15], [16]. And NDN allow the use of in-network caching and replication during the communication period, i.e., each piece of content can be cached on any intermediary node, thus decoupling the content from its provider in terms of location. In VNDN, vehicles can serve as consumers, providers and forwarders with caching capabilities, as well as RSU and intermediate forwarders with caching capabilities. Due to the mobility of vehicles, an important challenge is to address the backhaul link outage of data packets in VNDN. An effective solution is to predict the location of the vehicle's future access point through the motion information of vehicle and then forward the packet [17], [18].

Edge caching technology for vehicular networks has gained a lot of attention in recent years, where each roadside unit (RSU) can serve as an edge cache node, selectively storing content and distributing it to vehicles. However, the cache resources of RSUs are limited, and with the increasing amount of transmitted data, only using the cache resources of RSUs can no longer meet the needs of vehicle users. At the same time, the coverage of RSUs is also limited, and there are often discontinuous connections when vehicles pass through different RSUs. Therefore, vehicles are considered as mobile cache nodes, which on the one hand can relieve the pressure of insufficient RSUs cache resources, and on the other hand, vehicles cooperative cache can realize seamless

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connection within the RSUs interleaving coverage. Cooperative caching with vehicles is beneficial and reasonable compared to separate RSUs caches [19]. In addition, infotainment applications always have different request probabilities for their content [7]. Therefore, a more precise content caching strategy is needed, i.e., to determine whether specific content is cached at a particular network node (RSU or vehicle). With this caching strategy, users can directly access the desired content at a closer RSU or vehicle, significantly reducing content retrieval delay and transmission costs. The limited storage capacity of vehicles and RSUs is fully utilized to improve network performance in terms of average content delivery delay, average content delivery cost, and cache hit ratio.

In vehicular networks, both vehicles and RSUs can serve as content caching nodes, meaning that each vehicle can act as both a user and an edge caching node [20]. Therefore, all caching nodes can be divided into two categories: fixed roadside cache nodes (RCNs) and mobile vehicular cache nodes (VCNs) [21]. Existing caching strategies mostly consider vehicles as edge cache nodes but overlook the impact of frequently changing network topologies caused by the mobility of the vehicle on the caching strategy, and when the network topology changes, the caching performance will be greatly reduced. Furthermore, the differences between RCN and VCN, as well as the cooperation among cache nodes, have not been addressed in the existing literature. Therefore, in this paper, we propose an adaptive cooperative caching (ACC) strategy that combines RSUs and vehicles, which not only adapts to the effects of frequently changing network topology on caching strategy, but also considers the differences between VCNs and RCNs and the cooperation between cache nodes. The primary objective of this paper is to fully utilize the cache capacity of RSUs and vehicles by designing different caching strategies for the RCN layer and VCN layer, with the aim of minimizing overall transmission delay and communication overhead, and improving cache hit rates. First, we formulated an optimization problem to minimize transmission delay, this optimization considers not only content popularity but also network topology changes resulting from vehicle mobility. Then, addressing the differences between RCN and VCN layers, we divide the original problem into two sub-problems and solve them separately to obtain an approximate optimal solution for the best caching strategy. Simulation results demonstrate that our proposed ACC strategy outperforms existing caching strategies. The contribution of this paper is as follows:

- We first propose a three-layer edge cooperative caching network model, including gNB layer, RCN layer and VCN layer. In this network model, gNB is responsible for packet distribution, RCN layer and VCN layer respectively adopt different caching strategies, and cache nodes cooperate with each other to complete the cache task.
- In addition, for the problem of content popularity update and frequent changes of network topology due to vehicle mobility, two-slot model is proposed which is more suitable for real scenarios and improves the reliability of the caching strategy. In the two-time slot model, there are large time slots and small time slots, which represent the content popularity update time slots and network topology update time slots, respectively.
- Then, based on the above model, we formulate an optimization problem to minimize transmission delay, and propose an ACC strategy that can adapt to the changes of network topology by predicting the network topology under different time slots. Otherwise, considering the different characteristics of RCN layer and VCN layer, the original optimization problem is divided into two different sub-problems, the RCN layer optimization problem is transformed into MCK problem, and the VCN layer optimization problem is transformed into MMWDS problem.
- We propose two low-complexity greedy algorithms to solve the MCK problem and MMWDS problem, respectively, to obtain the near-optimal solution of the original optimization problem.
- Finally, we conduct extensive simulations to evaluate the performance of the proposed ACC strategy. The simulation results show that the proposed ACC strategy can effectively reduce the transmission delay and communication overhead, and improve the cache hit ratio.

The rest of this paper is organized as follows. Section II introduces the related work. Section III describes the system model and formulates the optimization problem. Section IV proposes the ACC strategy. Section V presents the simulation results. Finally, Section VI concludes the paper.

II. Related Work

Several research studies have been conducted to improve caching efficiency by considering different aspects such as content popularity prediction, mobility of vehicles and cache notes, and clustering methods.

Some caching strategies aim to predict the popularity of content to improve caching efficiency. For example, in [22], user requests are modeled by local and global Markov processing, and a reinforcement learning (RL) framework was proposed to find the optimal caching policy when the involved transition probabilities are unknown. In addition, an edge cooperative caching (ECC) based on machine learning and the greedy algorithm proposed in [23]. The neural cooperative filtering was used to design the content popularity prediction algorithm, and then the optimized greedy algorithm was used to obtain the content distribution strategy of various servers in the cooperative caching domain. A dynamic probabilistic caching was designed in [24] for scenarios where instantaneous content popularity may vary over time. They predict the average content popularity over a time window to optimize caching decisions. [25] studied the dynamic process of caching in vector space and introduces the concept of state transition field (STF) to model and characterize replacement strategies. Based on the edge computing concept, an efficient and privacy-preserving data downloading scheme for VANETs was proposed in [6], where RSU caching popular data in nearby qualified vehicles called edge computing vehicles (ECVs), and if vehicles wish to download popular data, they can directly download it from nearby ECVs. However, most of these works consider the spatio-temporal dynamics of content

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popularity and ignore the impact of the mobility of content requesters on the availability of cached content, especially in a highly dynamic environment such as VANET.

Meanwhile, some caching schemes applied to VANETs consider the mobility of vehicles. For instance, a vehicle mobility prediction module was proposed in [7] to estimate future connected RSUs. They designed a multi-layer caching mechanism that employs an over-the-top content popularity estimation scheme to predict the content request distribution. In [26], the benefits of proactive caching in RSUs are studied, taking into account the impact of vehicle velocity on optimal caching decisions. In the context of content delivery for parked vehicles, [27] proposes a content delivery framework where mobile vehicles can obtain content from RSUs and parked vehicles through competition and cooperation. A predictive proactive caching framework (PPCF) was proposed in [28], utilizing a long short-term memory network trained by particle swarm optimization (PSO-LSTM) to make accurate travel time predictions. In [29], a learning-based edge caching mechanism was proposed, which designs a new learning-based cooperative caching strategy (LECS) to evaluate the value of all the content, by cooperating between edge servers to maximize the value of the system. However, these works mainly consider RSUs or parked vehicles as cache nodes, neglecting the potential use of moving vehicles as cache nodes.

Therefore, some works consider vehicles with mobility as cache nodes. For instance, [30] proposed an NDN-based distributed probabilistic caching (DPC) strategy for VANET, in which each node makes caching decisions individually and independently based on content popularity and local cache state. A well-designed framework called IV-Cache to facilitate vehicular caching in vehicular scenarios was proposed in [31]. They proposed a caching scheme that enhances vehicle mobility through dynamic relaying among vehicles, with special consideration for vehicle mobility and vehicle-to-vehicle (V2V) communication vulnerabilities. In [32], the caching nodes that share more social attributes with content requesters are clustered into one group, and a vehicular content centric network (VCCN) cooperative caching scheme based on social attributes and mobility prediction (CCSAMP) is proposed. A dynamic pricing-based incentive mechanism was proposed in [33], which combined content information and link state to stimulate content sharing in C-V2X-based vehicular networks. For in-vehicle networks, [34] proposed a cooperative caching strategy, and then proposed a deep reinforcement learning (DRL) -based cooperative content caching scheme to maximize the data throughput obtained from edge devices. A two-tier distributed content caching scheme for VANETs was proposed in [8], to reduce the complexity, author propose a cooperation-based greedy (CBG) algorithm with linear complexity to solve the large-scale caching placement problem. However, most of these works consider caching independently and do not consider cooperative caching among caching nodes.

At the same time, some works use clustering methods to improve caching efficiency. For example, [35] discussed the time-delay optimal cooperative edge caching in large-scale user-centered mobile networks, and optimized content placement and cluster size based on random information such as network topology, traffic distribution, channel quality, and content popularity. In [36], a novel approach was proposed to support cooperative caching in disruption tolerant networks. Data was cached in a set of network centric locations (NCLs), which can share and coordinate cached data among multiple nodes, thereby reducing data access delay. A cluster-based cooperative caching approach with mobility prediction (COMP) was proposed in [11] for VNDN, where vehicles with similar mobility patterns are grouped by mobility prediction, and then a collaborative caching is proposed to structure intra- and inter-cluster communication of vehicles to mitigate the effects of vehicle mobility. In [37], a cluster-based content caching (CCC) architecture was proposed, which combines the advantages of distributed caching and centralized signal processing. A peer-to-peer cooperative caching mechanism among vehicles in urban scenarios was proposed in [38]. They also proposed a cooperative gateway cache invalidation (CGCI) mechanism based on cooperative caching invalidation technology. A novel cooperative caching scheme using mobility prediction and consistent hash for VCCN (CCMPCH) is designed in [39], which adopted partial matching prediction (PPM) to predict the path of each vehicle, clusters vehicles with similar future paths, moving directions, and velocities into a group, and then used the consistent hashing algorithm within the cluster to distribute content among cooperative nodes. However, these works ignore the difference between the mobility of vehicular cache nodes and the fixity of edge cache nodes, so the caching strategies proposed by the above works still have limitations.

To sum up, the existing caching strategies have their limitations. They do not consider the impact of frequently changing network topology on the caching strategy, as well as the differences and cooperation between fixed edge cache nodes and moving vehicle cache nodes. Therefore, to solve the above problems, the ACC strategy proposed in this paper can adapt to the changes of network topology and adopt different cooperative caching strategies according to the differences between fixed cache node (RCN) and mobile cache node (VCN) to minimize transmission delay.

III. Systems Model and Problem Formulation

In this section, we will build a system model, including a scenario and communication model, a vehicle mobility model, a two-slot model, a content popularity model, and a caching model. The terminologies defined in this section is listed as Table I.

A. Scenario and Communication Model

We consider a scenario of the most common two-way road in urban roads as shown in Fig. 1. The scenario consists of three main network layers of architecture: 1) the topmost next generation node B (gNB) connected to the cloud network, 2) the RCN layer consisting of multiple RCN placed along the road, and 3) the VCN layer consisting of all vehicle users on the current road segment. We assume that the current road segment is within the communication range of gNB, which is connected to the

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TABLE I
MAIN PARAMETERS IN THIS PAPER

Notation	*Semantics*
N^r	Number of RCNs
N^v	Number of VCNs
N^f	Number of contents
N^t	Number of small slots in a large slot
t	Topology update time slot
t^u	Content popularity update time slot
v_i	The i-th VCN
r_j	The j-th RCN
f_k	The k-th content
S^{r}	Cache capacity of RCNs
S^{v}	Cache capacity of VCNs
S^{f}	Size of contents
S^p	Size of the packet
u_i	Velocity of the i-th vehicle
P_k^f	Popularity of the k-th content
r^{vv}	Transmission rate of V2V
r^{vi}	Transmission rate of V2I
r^{vn}	Transmission rate of V2N
r^w	Transmission rate of wire
$\mathcal{M}^{\mathcal{V}}$	Cache placement mapping matrix for VCNs
$\mathcal{M}^{\mathcal{R}}$	Cache placement mapping matrix for RCNs
D^{avg}	Average transmission delay
B	Caching benefit

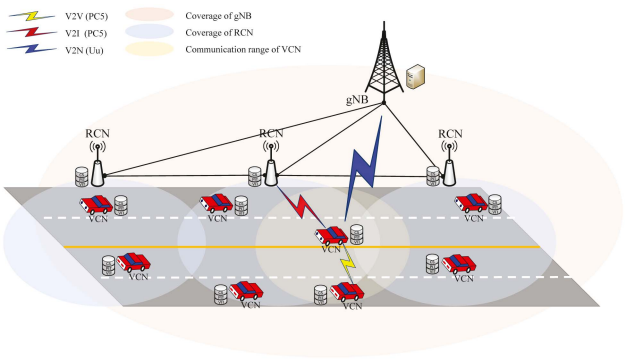


Fig. 1. Three-layer network architecture.

cloud network via optical fiber. The gNB can fetch all the content from the cloud and predict their content popularity within the coverage [8]. In the RCN layer, all the RCNs are placed along the road, all RCNs form a set R=$\{r_1, r_2, \ldots, r_{N^r}\}$, where N^r denotes the number of RCNs on the roadway. In the VCN layer, each vehicle can be both a content requester and a content caching node, and all VCN form the set V=$\{v_1, v_2, \ldots, v_{N^v}\}$, where N^v denotes the number of VCNs on the roadway.

In this paper, we instantiate VNDN as the underlying network architecture of the whole communication system. The communication under the VNDN architecture adopts the pull model for content propagation, in which the vehicle first sends the interest containing the required content name and broadcasts it to its neighbors. Interest can be stored, carried, and forwarded by the vehicle or RSU for a maximum time-to-live (TTL) second until it reaches the content of interest [39]. Each side unit acts as a gateway, responsible for forwarding interest to other side units in the event of local unavailability. The gNB is connected to the cloud network and acts as the source publisher of all the data, and when all the vehicles and RSUs on the road do not cache the data content matching the interest, the interested requestor will get the content from the gNB. Each vehicle and RSU maintains three data structures, including a content store (CS), a pending interest table (PIT) and a vehicle information table (VIT). CS is used to cache popular content, PIT is used to track forwarded interest, and VIT is used to record its neighbor information. When the node receives an interest request, it first searches for the requested content name in its CS, returns the packet if there is a hit, otherwise it adds the content name of the interest to the PIT.

In addition, each vehicle uses C-V2X communication technology with an On-Board Unit (OBU), C-V2X provides the integration of both Device-to-Device (D2D) communication based on the PC5 interface, and cellular communication based on the Uu interface [40]. There are five types of communication modes involved in this paper, namely VCN-to-VCN (i.e., V2V), VCN-to-RCN (i.e., V2I), VCN-to-gNB (i.e., V2N), RCN-to-RCN (i.e., I2I), and RCN-to-gNB (i.e., I2N). V2V, V2I, and V2N are wireless links that use C-V2X communication technology, where V2V, V2I use D2D communication based on PC5 interfaces, and V2N uses cellular communication based on Uu interfaces. I2I, I2N use wired fiber optic communication. It is worth mentioning that in 5 G NR-V2X, both PC5 interface and Uu interface support unicast, multicast and broadcast transmission modes [41]. Furthermore, since all VCNs and RCNs are under gNB coverage, gNB allocates radio resources for all D2D communications, and all D2D communications adopt orthogonal frequency division multiplexing (OFDM) technology, so interference is not considered in the communication model [42].

Therefore, we can obtain the transmission rate r^{vv} for V2V:

$$r^{vv} = B^{vv} \log_2 \left(1 + \frac{P^{vv} g^{vv}(d^{vv})}{(\sigma^{vv})^2}\right), \tag{1}$$

where B^{vv} is the bandwidth of the V2V communication channel, P^{vv} is the transmit power of the vehicle, $g^{vv}(d^{vv})$ is the path loss of the V2V communication channel, d^{vv} is the distance between the two vehicles,and σ^{vv} is the noise power of the V2V communication channel. In addition, the path loss of the V2V communication channel is calculated by the following equation:

$$g^{vv}(d^{vv}) = \frac{\alpha^{vv}}{(d^{vv})^{\beta^{vv}}}, \tag{2}$$

where α^{vv} and β^{vv} are the path loss parameters [43]. Similarly, we can obtain the transmission rate r^{vi} for V2I and the transmission rate r^{vn} for V2N according to the above formula. As for the wired transmission rate r^w between RCNs and between RCN and gNB, it is a fixed constant.

B. Vehicle Mobility Model

The velocities of all vehicles on the road at time slot t can be denoted as $\mathrm{U}(t) = \{u_1(t), u_2(t), \ldots, u_{N^v}(t)\}$, and $u_i(t)$ denotes the total number of vehicles on the road. We assume that during the initialization phase (i.e., $t = 0$), the velocity $u_i(t)$ of vehicle v_i satisfies a truncated Gaussian distribution with maximum value u_{max}, minimum value u_{min}, mean value $\frac{u_{max}+u_{min}}{2}$ and variance 0.5, where u_{min} and u_{max} are the

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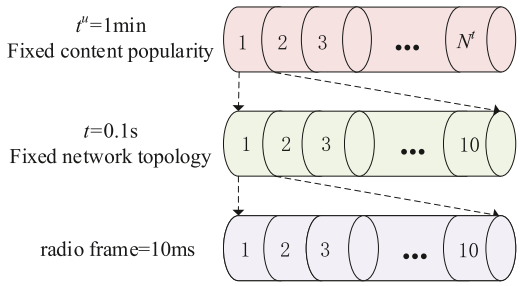


Fig. 2. Two-time slot model.

minimum and maximum velocities of the vehicles, respectively [8]. Without loss of generality, the velocity $u_i(t)$ of vehicle i $(i \in 1, 2, \ldots, u_{N^v}(t))$ at time slot t is highly correlated with the velocity $u_i(t-1)$ at the previous moment, and $u_i(t)$ approximately obeys a truncated Gaussian distribution with mean $u_i(t-1)$ and variance 0.5 [43], then the probability density function of velocity u_i at time slot t can be expressed as:

$$f(u_i(t)) = \begin{cases} \frac{\phi(\frac{u_i(t)-\mu}{\sigma})}{\sigma(\Phi(\frac{u_{\max}-\mu}{\sigma})-\Phi(\frac{u_{\min}-\mu}{\sigma}))}, \\ \qquad u_{\min} \le u_i(t) \le u_{\max} \\ 0, \qquad \text{otherwise} \end{cases}, \tag{3}$$

where $\mu = u_i(t-1)$ and $\sigma = 0.5$ are the mean and standard deviation of the vehicle velocity distribution, respectively. And, $\phi(x)$ is the probability density function of the standard normal distribution, $\Phi(x)$ is the cumulative distribution function of the standard normal distribution, which can be expressed as:

$$\phi(x) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}x^2\right), \tag{4}$$

$$\Phi(x) = \frac{1}{2}\left(1 + erf\left(\frac{x}{\sqrt{2}}\right)\right). \tag{5}$$

Therefore, the probability density function of the velocity $\mathrm{U}(t+1)$ of all vehicles at time slot t+1 can be obtained from the velocity $\mathrm{U}(t)$ of vehicles at the current time slot t. Finally, the velocity of the vehicle at slot t+1 can be simply predicted by random sampling.

C. Two-Time Slot Model

Without loss of generality, the content popularity is updated much less frequently than the vehicle network topology and the transmission rate due to the mobility of the vehicle [44]. Therefore, to better reflect the characteristics of the real scene, we propose a two-slot model, as shown in Fig. 2, where Nt denotes the number of small slots in a large slot. In the two-slot model, there are large time slot t^u and small time slot t, and we define the small time slot t as the duration of 10 radio frames, usually 0.1 s, while the large time slot t^u is defined as the update frequency of the content popularity, usually 1 min.

Since a small time slot t is only 0.1 s, the velocity of the vehicle on the urban road is generally 8-12 m/s, then the distance of all vehicles moving within 0.1 s is only about 1 m. The communication range of the vehicle is 100 m, so it can be considered that the network topology of the VCN layer is almost constant in a small time slot t. Therefore, in this paper we assume that in a small time slot t, the network topology and transmission rate remain unchanged, while in a large time slot t^u, the popularity of content remains unchanged. At the same time, gNB delivers the content that needs to be cached to a different cache node within each small time slot, updating the popularity of everything at the start of the next large time slot.

D. Content Popularity Model

The set of all contents cached in the gNB is denoted as F= $\{f_1, f_2, \ldots, f_{N^f}\}$, and N^f denotes the total number of contents in the gNB. In the initialization phase, considering that some contents may have higher hotness than others, we assume that the popularity of the contents in the gNB obeys the Zipf distribution [45], [46], so we can obtain the initialization content popularity P_k^f for the k-th content f_k as follows:

$$P_k^f = \frac{k^{-\gamma}}{\sum_{j=1}^{N^f} j^{-\gamma}}, \tag{6}$$

where γ is the parameter of the Zipf distribution. After that, the content popularity needs to be updated every time slot t^u. Without loss of generality, the content popularity of the next time slot t^u+1 is correlated with the frequency of vehicles accessing content within the current time slot t^u and also has some randomness.

E. Caching Model

In our proposed ACC strategy, the RCN layer and VCN layer use centralized and distributed caching strategies respectively. Therefore, We define two two-dimensional matrices $\mathcal{M}^{\mathcal{V}}$ and $\mathcal{M}^{\mathcal{R}}$ to represent the cache placement mapping of VCNs and RCNs in a large time slot t^u, respectively, which can be expressed as:

$$\mathcal{M}^{\mathcal{V}} = \begin{bmatrix} m^v_{1,1} & m^v_{1,2} & \cdots & m^v_{1,N^f} \\ m^v_{2,1} & m^v_{2,2} & \cdots & m^v_{2,N^f} \\ \vdots & \vdots & \ddots & \vdots \\ m^v_{N^v,1} & m^v_{N^v,2} & \cdots & m^v_{N^v,N^f} \end{bmatrix}, \tag{7}$$

$$\mathcal{M}^{\mathcal{R}} = \begin{bmatrix} m^r_{1,1} & m^r_{1,2} & \cdots & m^r_{1,N^f} \\ m^r_{2,1} & m^r_{2,2} & \cdots & m^r_{2,N^f} \\ \vdots & \vdots & \ddots & \vdots \\ m^r_{N^r,1} & m^r_{N^r,2} & \cdots & m^r_{N^r,N^f} \end{bmatrix}, \tag{8}$$

where N^v is the number of VCN on the road, N^r is the number of RCNs on the road, and N^f is the number of contents in the gNB.

The elements of the matrix $\mathcal{M}^{\mathcal{V}}$ and $\mathcal{M}^{\mathcal{R}}$ are binary variables, and the value of 1 indicates that the corresponding vehicle or RCN caches the corresponding content within time slot t^u, and the value of 0 indicates that the corresponding vehicle or RCN does not cache the corresponding content within time slot t^u. For example, the element $m^v_{2,3}$=1 indicates that the second vehicular caches the third content within time slot t^u. Similarly, the element $m^r_{2,3}$=0 indicates that the second RCN

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does not cache the third content within the time slot t^u. It is worth mentioning that whenever the popularity is updated, all caching nodes only need to replace some of the cached content without re-downloading all the content.

F. Problem Formulation

In this section, we formulate the problem as an objective optimization problem, the objective function of the problem is to minimize the average transmission delay for all vehicle users on the road to access contents in a large time slot t^u, and the constraints are the cache capacity of the RCN and the cache capacity of the VCN. The size of all the content in the gNB can be expressed as $\mathrm{S^f}=\{s_1^f, s_2^f, \ldots, s_{N^f}^f\}$, the cache capacity is the same for all RCNs and all VCNs, and can be denoted as $\mathrm{S^r}=\{s_1^r, s_2^r, \ldots, s_{N^r}^r\}$ and $\mathrm{S^v}=\{s_1^v, s_2^v, \ldots, s_{N^v}^v\}$. We define the time for a vehicle user to receive a complete packet as the transmission delay D. Then the transmission delay $D_{i,k}$ for vehicle v_i to download content f_k can be expressed as:

$$
\begin{aligned}
D_{i,k} &= \begin{cases} 0, & \text{if } C_0 \\ \frac{S^p}{r_{i,k}^{vv}}, & \text{if } C_1 \\ \frac{S^p}{r_{i,k}^{vi}}, & \text{if } C_2 \\ \frac{S^p}{r_{i,k}^{vi}} + \frac{S^p}{r_{i,k}^{w}}, & \text{if } C_3 \\ \frac{S^p}{r_{i,k}^{vn}}, & \text{otherwise} \end{cases} \\
C_0 &: m_{i,k}^v \neq 0, \\
C_1 &: \left(m_{i,k}^v = 0\right) \& \left(\sum_{i=1}^{N^r} m_{i,k}^v \neq 0\right), \\
C_2 &: \left(\sum_{i=1}^{N^r} m_{i,k}^v = 0\right) \& \left(m_{j,k}^r \neq 0\right), \\
C_3 &: \left(\sum_{i=1}^{N^r} m_{i,k}^v = 0\right) \& \left(m_{j,k}^r = 0\right) \& \\
&\quad \left(\sum_{j=1}^{N^r} m_{j,k}^r \neq 0\right),
\end{aligned} \tag{9}
$$

where S^p denotes the size of the packet, C_0 indicates that the content f_k requested by the vehicle user v_i is cached exactly locally, C_1 indicates that the content f_k requested by the vehicle user v_i is cached on a nearby vehicle that can communicate; C_2 indicates that the content f_k requested by the vehicle user v_i is cached on the RCN r_j and the vehicle user v_i is within the coverage of the RCN r_j; C_3 indicates that the content f_k requested by the vehicle user v_i is cached on other RCN than r_j; otherwise indicates that the content f_k requested by the vehicle user v_i is not cached at the edge node and needs to be sent to the gNB to obtain the content.

Therefore, the average transmission delay D^{avg} for all vehicle users on the road to access contents can be expressed as:

$$
D^{avg} = \frac{1}{N^v} \sum_{i=1}^{N^v} \sum_{k=1}^{N^f} P_k^f D_{i,k}, \tag{10}
$$

where N^v and N^f denote the number of VCN and the number of content, respectively.

Therefore, the objective function of the minimize average transmission delay problem P_1 can be expressed as follows:

$$
\begin{aligned}
&\min D^{avg} \\
&s.t. \sum_{k=1}^{N^f} m_{j,k}^r s_k^f \leq s_j^r, j \in \{1, 2, \ldots, N^r\}, \\
&\quad \sum_{k=1}^{N^f} m_{i,k}^v s_k^f \leq s_i^v, i \in \{1, 2, \ldots, N^v\}, \\
&\quad m_{j,k}^r \in \{0, 1\}, m_{i,k}^v \in \{0, 1\},
\end{aligned} \tag{11}
$$

where $m_{j,k}^r$ and $m_{i,k}^v$ denote the caching decision of the RCN r_j and the VCN v_i for the content f_k, respectively. s_k^f denote the size of content f_k. The first constraint indicates that the total size of the cached content in the RCN r_j should be less than the cache capacity s_j^r, and the second constraint indicates that the total size of the cached content in the VCN v_i should be less than the cache capacity s_i^v. The third constraint indicates that the caching decision of the RCN r_j and the VCN v_i can only be 0 or 1, which means that the content f_k is cached or not cached.

P_1 is a 0-1 integer programming problem and solving this optimization problem is usually NP-hard [8]. Otherwise, considering that RCNs are stationary and connected to gNB through a wired connection with a fast and stable transmission rate, while VCNs have fast mobility, resulting in unstable link connections to gNB and relatively slow transmission rates, different caching policies need to be designed for the RCN and VCN layers. Thus, to reduce the complexity of P_1, we decompose the P_1 into two relatively simple subproblems P_2 and P_3, and solve them separately.

P_2 is described as solving for the optimal $\mathcal{M}^{\mathcal{R}}$ such that the average transmission delay is minimized when $\mathcal{M}^{\mathcal{V}}$ has been determined, where $\mathcal{M}^{\mathcal{R}}$ and $\mathcal{M}^{\mathcal{V}}$ denote the cache placement mapping of RCNs and VCNs, respectively. Therefore, the objective function of P_2 can be expressed as:

$$
\begin{aligned}
&\arg\min D^{avg}(\mathcal{M}^{\mathcal{R}}) \\
&s.t. \sum_{k=1}^{N^f} m_{j,k}^r s_k^f \leq s_j^r, j \in \{1, 2, \ldots, N^r\}, \\
&\quad m_{j,k}^r \in \{0, 1\}.
\end{aligned} \tag{12}
$$

Similarly, P_3 is described as solving for the optimal $\mathcal{M}^{\mathcal{V}}$ such that the average transmission delay is minimum when $\mathcal{M}^{\mathcal{R}}$ has been determined, the objective function of P_3 can be expressed as:

$$
\begin{aligned}
&\arg\min D^{avg}(\mathcal{M}^{\mathcal{V}}) \\
&s.t. \sum_{k=1}^{N^f} m_{i,k}^v s_k^f \leq s_i^v, i \in \{1, 2, \ldots, N^v\}, \\
&\quad m_{i,k}^v \in \{0, 1\}.
\end{aligned} \tag{13}
$$

Although P_2 and P_3 are greatly simplified compared to P_1, their objective functions are still very complex. To simplify the objective functions of P_2 and P_3, we propose an ACC strategy, which is elaborated in the next section.

IV. Adaptive Cooperative Caching (ACC) Strategy

This section proposes an ACC strategy for vehicular networks. The proposed ACC strategy consists of the RCN layer with

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centralized caching decisions and the VCN layer with distributed caching decisions. In detail, We will introduce the RCN and VCN layers' caching strategies in the following.

A. RCNs Caching Strategy

In this section, we propose a centralized caching strategy for RCNs. In the RCN layer, RCNs actively upload their caching status, remaining capacity, and environmental information to the gNB at the beginning of time slot t^u. Based on all the information uploaded by the RCN, gNB needs to calculate the caching benefits of all the contents and then make a caching decision.

Without loss of generality, for any content, the content with high content popularity should have a higher caching benefit. However, the property describing a content f_k includes the size s_k^f and the popularity P_k^f. In addition, we also consider the effect of the number of content f_k caches in RCNs on the caching benefit, since content with higher popularity may be duplicated by different RCNs. Thus, the higher the popularity P_k^f, the smaller the size s_k^f, and the smaller the number of copies N^k caches in the RCN layer, the greater the caching benefit from caching content f_k (i.e., the average transmission delay of cached content f_k is lower than the average transmission delay of cached other content).

To evaluate the caching benefit of different copies of different contents cached in different RCN, a caching benefit evaluation criterion is designed to evaluate the caching benefit of the N^k-th copy of content f_k cached in j-th RCN r_j. Thus, the caching benefit $B_{j,k}(N^k)$ can be calculated as:

$$B_{j,k}(N^k) = \frac{P_k^f}{s_k^f} D_{j,k,N^k}^{avg}, \tag{14}$$

where N_j^v denotes the number of vehicle users under the coverage of the j-th RCN, $0 < N^k \leq N^r$ since each RCN does not allow duplicate content to be cached. D_{j,k,N^k}^{avg} represent the reduction of average transmission delay for all vehicles on the road to access content f_k with the N^k-th copy cached in j-th RCN r_j when the number of copies of content in the RCN layer is $N^k - 1$, it can be calculated as:

$$D_{j,k,N^k}^{avg} = D_{k,N^k-1}^{avg} - D_{k,N^k}^{avg}, \tag{15}$$

D_{k,N^k}^{avg} represent the average transmission delay for all vehicles to access content f_k with N^k copies, it can be calculated as:

$$D_{k,N^k}^{avg} = \frac{1}{N^v}\left(\sum_{j=1}^{N^r}\sum_{i=1}^{N_j^v} D_{i,j,k}^{vi} + \sum_{j=N^k+1}^{N^r}\sum_{i=1}^{N_j^v} D_{i,j,k}^{w}\right), \tag{16}$$

where $D_{i,j,k}^{vi} = \frac{S^p}{r_{i,j,k}^{vi}}$ denotes the transmission delay of the i-th vehicle user under the coverage of the j-th RCN r_j to obtain the content f_k from r_j, and $D_{i,j,k}^{w} = \frac{S^p}{r_{i,j,k}^{w}}$ denotes the transmission delay of the j-th RCN r_j to obtain the content f_k from other RCNs.

At the same time, when no content f_k is cached in the RCN layer (i.e., N^k=0), the average transmission delay can be computed as:

$$D_{k,0}^{avg} = \frac{1}{N^v}\sum_{i=1}^{N^v} D_{i,k}^{vn}, \tag{17}$$

where $D_{i,k}^{vn}$ denotes the transmission delay of the i-th vehicle user to access the content f_k from the gNB. Therefore, we can get the caching benefit $B_{j,k}(N^k)$ as follows:

$$B_{j,k}(N^k) = \frac{P_k^f}{s_k^f} D_{j,k,N^k}^{avg} = \begin{cases} \frac{P_k^f}{s_k^f N^v}\left(\sum_{i=1}^{N^v} D_{i,k}^{vn} - \sum_{j=1}^{N^r}\sum_{i=1}^{N_j^v}(D_{i,j,k}^{vi} + D_{i,j,k}^{w}) + \sum_{i=1}^{N_j^v} D_{i,j,k}^{w}\right), & \text{if } N^k = 1 \\ \frac{P_k^f}{s_k^f N^v}\sum_{i=1}^{N_j^v} D_{i,j,k}^{w}, & \text{otherwise.} \end{cases} \tag{18}$$

Algorithm 1: Cache Placement Algorithm for RCNs.

Input: $N^f, N^r, \mathrm{P^f}, \mathrm{S^f}$
Initialize: $\mathcal{M}^{\mathcal{B}}$, $\mathcal{M}^{\mathcal{R}}$, $\mathrm{C^p}$, and $\mathrm{S^r}$
Calculate the number of vehicle users under the coverage of all RCNs and sort them in descending order to get R
Calculate the content popularity per unit size and sort them in descending order to get F
Cache the first content in the first RCN and update $\mathcal{M}^{\mathcal{R}}$ and $\mathrm{S^r}$
calculate the values of $m_{1,2,2}^b$ and $m_{2,1,1}^b$, and add them to the $\mathrm{C^p}$
While $\mathrm{C^p} \neq \emptyset$ **do**
  Assign the coordinates of the maximum value in $\mathrm{C^p}$ to k, N^k, and j, and then remove it from $\mathrm{C^p}$
  Update $\mathcal{M}^{\mathcal{R}}$ and $\mathrm{S^r}$:
    $m_{j,k}^r \Leftarrow 1$
    $s_j^r \Leftarrow s_j^r - s_k^f$
  if $m_{k-1,N^k+1,j+1}^b \notin \mathrm{C^p}$
    for $x = j+1$ to N^r **do**
      if $s_x^r \geq s_k^f$
        Calculate $m_{k,N^k+1,x}^b$ and add it to $\mathrm{C^p}$
        break
      end if
    end for
  end if
  if $m_{k+1,N^k-1,j-1}^b \notin \mathrm{C^p}$
    for $x = j$ to N^r **do**
      if $s_x^r \geq s_{k+1}^f$
        Calculate $m_{k+1,N^k,x}^b$ and add it to $\mathrm{C^p}$
        break
      end if
    end for
  end if
end while
Output: $\mathcal{M}^{\mathcal{R}}$

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Hence, the average transmission delay of the system is minimized when the sum of the cache gains of all contents cached in the RCN layer is maximized. The optimization problem P_2 can be transformed into the optimization problem P_4 as follows:

$$\begin{aligned} &\max \sum_{j=1}^{N^r}\sum_{k=1}^{N_j^f} m_{j,k}^r B_{j,k} \\ &s.t. \sum_{k=1}^{N^f} m_{j,k}^r s_k^f \le s_j^r, j \in \{1,2,\ldots,N^r\}, \\ &m_{j,k}^r \in \{0,1\}, \end{aligned} \tag{19}$$

where N_j^f is the number of contents cached in the j-th RCN, and $m_{j,k}^r$ is the binary variable indicating whether the k-th content is cached in the j-th RCN. The optimization problem P_4 is a MCK problem [47], where each content f_k has N^k copies, with size s_k^f, value $B_{j,k}$, and the knapsack capacity is s_j^r. Therefore, we can use the dynamic programming algorithm to solve the problem P_4. However, since the solution of problem P_4 requires calculating all possible combinations of all copies of all contents, it will lead to very high computational complexity. Therefore, we propose an approximate solution based on the greedy algorithm to reduce the computational complexity.

In the initialization phase, a $N^f \times N^r \times N^r$ three-dimensional matrix $\mathcal{M}^{\mathcal{B}}$ is constructed to represent the caching benefits of different copies of the content cached on different RCNs, then any element $m_{k,N^k,j}^b$ in $\mathcal{M}^{\mathcal{B}}$ represents the caching benefit of the N^k-th copy of the k-th content cached on the j-th RCN. Sorting all RCNs in descending order based on the number of vehicles under their coverage, so the number of vehicles under the coverageof all RCNs can be represented as $N_1^v > N_2^v > \ldots > N_{N^r}^v$. Meanwhile, the content popularity per unit size is calculated for all contents and they are sorted in descending order from largest to smallest, so there are $\frac{P_1^f}{s_1^f} > \frac{P_2^f}{s_2^f} > \ldots > \frac{P_{N^f}^f}{s_{N^f}^f}$.

Then for any content f_k and number of copies N^k, $\mathcal{M}^{\mathcal{B}}$ satisfies $m_{k,N^k,1}^b > m_{k,N^k,2}^b > \ldots > m_{k,N^k,N^r}^b$, for any copy number N^k and RCN r_j, $\mathcal{M}^{\mathcal{B}}$ satisfies $m_{1,N^k,j}^b > m_{2,N^k,j}^b > \ldots > m_{N^f,N^k,j}^b$, for any content f_k and RCN r_j, $\mathcal{M}^{\mathcal{B}}$ satisfies $m_{k,1,j}^b > m_{k,2,j}^b = \cdots = m_{k,N^r,j}^b$. It can be observed that the first copy of the first content cached in the first RCN corresponds to the caching benefit $m_{1,1,1}^b$ which is the maximum value in $\mathcal{M}^{\mathcal{B}}$.The specific algorithmic process is as shown in Algorithm 1.

It is worth mentioning that, in line 18 to line 21 of Algorithm 1, if $m_{k+1,N^k-1,j-1}^b$ is not in $\mathrm{C^p}$, it indicates that the (N^k-1)-th copy of the $(k+1)$-th content is already cached in the $(j-1)$-th RCN. Since the same content cannot be cached in the same RCN, we need to consider whether to cache the N^k-th copy of the $(k+1)$-th content in the next RCN. If the remaining cache capacity of RCN j is greater than the size of the $(k+1)$-th content, we need to calculate $m_{k+1,N^k,j}^b$ and add it to $\mathrm{C^p}$. If $m_{k+1,N^k-1,j-1}^b$ is already in $\mathrm{C^p}$, there is no need to calculate the caching benefit for the N^k-th copy of the $(k+1)$-th content because $m_{k+1,N^k-1,j-1}^b$ is guaranteed to be greater than $m_{k+1,N^k,j}^b$. In the worst case, the complexity of Algorithm 1 could be $\mathrm{O}(N^f \times N^r)$, where N^r is the number of RCNs and N^f is the number of contents.

Algorithm 2: Cache Placement Algorithm for VCNs.

Input: $N^f, N^r, \mathrm{P^f}, \mathrm{S^f}$
Initialize: $\mathcal{M}^{\mathcal{V}}$, and $\mathrm{S^v}$
Calculate the content popularity per unit size and sort them in descending order to get F
Predict the network topology for all small time slots to get G
for $k = 1$ to N^f **do**
  while G $\neq \emptyset$ **do**
    for all $v_i \in$ G **do**
      if $s_i^v \ge s_k^f$
        for t=1 to N^t **do**
          Calculate $W_{i,k}(t)$ of v_i according to Equation (23)
        end for
      end if
      Calculate $W_{i,k}$ of v_i according to (24)
    end for
    Add the vertex with the smallest $W_{i,k}$ value to the dominance subset
    $m_{i,k}^r \Leftarrow 1$
    $s_i^v \Leftarrow s_i^v - s_k^f$
    Remove v_i and its neighboring vertices from G
  end while
end for
Output: $\mathcal{M}^{\mathcal{V}}$

B. VCNs Caching Strategy

In this section, we propose a distributed caching strategy for VCNs. Unlike RCNs that are stationary and have stable link connections to gNB with fast transmission rates, the mobility of VCNs leads to frequent network topology changes, unstable link connections, and slow transmission rates. Therefore, in our proposed strategy, the VCN layer adopts a distributed cooperative caching model. That is, in the content caching phase, all VCNs calculate their cache weights in parallel according to the network topology and then upload them to gNB. Finally, gNB selects appropriate VCNs as the cache nodes of the content according to the cache weights of all VCNs. In order to minimize the average transmission delay with limited cache capacity, each vehicle user should obtain content from a local or neighboring VCN whenever possible. Therefore, for any content f_k, gNB needs to consider selecting the minimum number of VCNs to cache the content f_k so that any vehicle user can access the local or neighboring VCN to obtain the content f_k.

Network topology of the VCN layer in each time slot t can be equated to an undirected graph $G(t) = (V, E(t))$, as shown in Fig. 3. Where V denotes all the VCNs and $E(t)$ denotes

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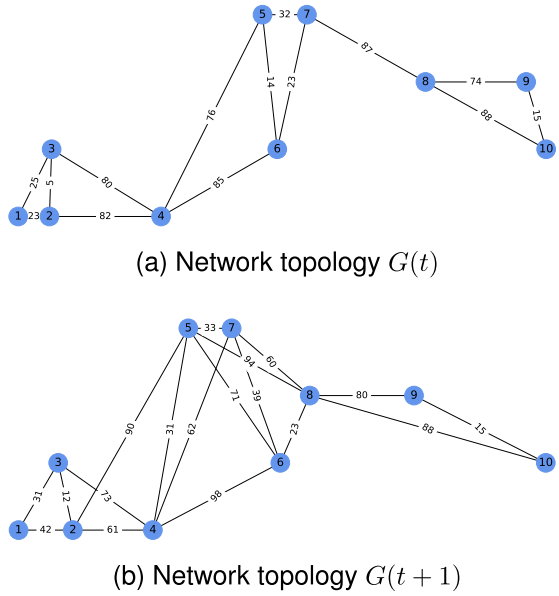


Fig. 3. Example of the network topology from $G(t)$ to $G(t+1)$.

the link connections between all the VCNs in time slot t that can communicate directly. In Fig. 3, there are 10 VCNs on the road, the edge between two VCNs indicates that they can communicate directly. And the edge weight represents the distance between the two VCNs, the longer the distance, the slower the transmission rate. When v_i's neighbors access the content f_k, they can download it directly from v_i, if the vertex v_i has cached the content f_k. For example, if v_4 has cached the content f_k, then v_1, v_2, v_3 and v_5 can download the content f_k from v_4 directly.

If v_i caches the content f_k, the average transmission delay $D_{i,k}^{avg}$ of each neighbor of v_i accessing content f_k can be computed as:

$$D_{i,k}^{avg} = \frac{\sum_{i'=1}^{N_i^v} D_{i,i',k}^{vv}}{N_i^v}, \tag{20}$$

where N_i^v denotes the number of neighbors of v_i, and $D_{i,i',k}^{vv} = \frac{S^p}{r_{i,i',k}^{vv}}$ denotes the transmission delay of $v_{i,i'}$ to obtain the content f_k from v_i, which can be calculated by (9). Assume that $D_{i,k,G(t)}^{avg}$ is taken as the vertex weight of v_i in $G(t)$, then when the network topology remains unchanged, the optimal caching placement problem with the minimum average transmission delay can be transformed into the minimum weighted dominating set (MWDS) problem. The MWDS problem can be described as follow: given an undirected graph $G = (V, E)$ with non-negative weights $D_{i,k,G(t)}^{avg}$ associated with each vertex v_i in V, the goal is to find a subset of V such that every vertex in V is either in the subset or adjacent to some vertex in the subset, and the sum of the weights of the vertices in the subset is minimized [48].

However, the topology of the network may change at each time slot due to the mobility of vehicles. Fig. 3(a) shows the network topology $G(t)$ at time slot t, where $\{v_5, v_7, v_8, v_9\}$ move to left and $\{v_1, v_2, v_3, v_4, v_6, v_{10}\}$ move to the right. Fig. 3(b) shows the network topology $G(t+1)$ at time slot $t+1$. In Fig. 3(a), there are three neighbors of v_5, however, in Fig. 3(b) there are five neighbors of v_5. Therefore, the optimal cache node of f_k in time slot t may not be the optimal cache node of f_k in time slot $t+1$.

To optimize caching decisions, VCNs need to consider not only the network topology $G(t)$ for the current time slot t, but also the network topology $\{G(t+1), G(t+2), \ldots, G(N^t)\}$ for future time slots, where N^t denote the number of small slots in a large time slot. Since the network topology at time slot t depends on the positions of all VCNs at time slot t, at the same time, the positions $x(t)$ and velocities $u(t)$ of VCNs at the time slot t are only related to the positions $x(t-1)$ and velocities $u(t-1)$ of VCNs at the previous time slot $t-1$. Therefore, according to (1), the velocity $u(t)$ of all vehicles at time slot t can be predicted, and then the position $x(t)$ of each vehicle at time slot t can be predicted according to the vehicle motion direction, and then the network topology $G(t)$ at any time slot can be similarly predicted. We assume that all the VCNs travel along a straight line on the road, then the position $x_i(t)$ of the i-th VCN at time slot t can be expressed as:

$$x_i(t) = x_i(t-1) + u_i(t-1)\Delta t, \tag{21}$$

where Δt denotes the time interval between two-time slots and $u_i(t-1)$ denotes the velocity of the i-th VCN at time slot $t-1$. Based on this we can get the network topology $\mathrm{G} = \{G(1), G(2), \ldots, G(N^t)\}$ for all time slots in a content popularity period, and then calculate the MWDS problem for all of the network topology graphs G.

Thus, we can transform the problem P_3 into the multiple minimum weighted dominating set (MMWDS) problem P_5 as follows:

$$\begin{aligned} &\min \sum_{t=1}^{N^t} \sum_{k=1}^{N^k} \sum_{i=1}^{N^v} m_{i,k}^v D_{i,k,G(t)}^{avg} \\ &s.t.\ m_{i,k}^v + m_{i',k}^v \geq 1, \quad \forall (i,i') \in E, \\ &\sum_{k=1}^{N^f} m_{i,k}^v s_k^f \leq S^v, i \in \{1,2,\ldots,N^v\}, \\ &m_{i,k}^v \in \{0,1\}. \end{aligned} \tag{22}$$

where $m_{i,k}^v$ denotes whether the v_i caches the content f_k. The first constraint in (22) means that if the v_i and the $v_{i'}$ are neighbors, then at least one of them should cache the content f_k. The second constraint means that the total size of the cached contents of v_i should not exceed the caching size s_i^v of the v_i. The third constraint means that the value of $m_{i,k}^v$ should be either 0 or 1.

This problem is known to be NP-hard, which means that no known polynomial-time algorithm can solve it exactly for all inputs. To solve the above problem, we use an approximation algorithm inspired by [49], [50]. It is mentioned that compared to considering only 1-hop weight, using 2-hop weight allows each node to gather more local information in [50]. Although it slightly increases the space complexity, it leads to an approximate improvement of approximately 11% in the final results. While increasing the hop count may yield approximation solutions closer to the optimal solution, it comes at the cost of rapidly increasing time and space complexity, as well as

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being constrained and affected by communication environments. Therefore, in this paper, we are using the 2-hop weight in Algorithm 2.

First, gNB needs to calculate the content popularity of all content under unit size and rank them in descending order to get F. It determines the priority of gNB deliver content, the higher the content popularity under unit size, the higher the deliver priority. At the same time, each VCN predicts its position in each small time slot within a large time slot according to (21), so that the network topology $G(t)$ under each small time slot can be obtained. Then each VCN calculates the ratio of the sum of the local weight and the neighbor weight respectively under all network topologies $G(t)$ as the 2-hop weight $W_{i,k}(t)$, as follows:

$$W_{i,k}(t) = \frac{D^{avg}_{i,k,G(t)}}{\sum_{i'=1}^{N_i^v} D^{avg}_{i,i',k,G(t)}}, \tag{23}$$

where i' denotes the number of neighbor vertices of v_i, $D^{avg}_{i,i',k,G(t)}$ denotes the weight of the i'th neighbor vertex $v_{i,i'}$ of v_i, and $D^{avg}_{i,k,G(t)}$ denotes the weight of v_i. Then, each VCN calculates the mean of the 2-hop weights $W_{i,k}(t)$ in all network topologies $G(t)$ to get $W_{i,k}$, as follows:

$$W_{i,k} = \frac{\sum_{t=1}^{N^t} \beta_t W_{i,k}(t)}{N^t}, \tag{24}$$

where β_t denotes the weight of the t-th network topology $G(t)$, $\sum_{t=1}^{N^t} \beta_t = 1$ and $\beta_1 > \beta_2 > \ldots > \beta_N^t$. The smaller $W_{i,k}$ indicates the smaller overall transmission delay when the content f_k is cached on VCN v_i. Therefore, the smaller the value of $W_{i,k}$ the higher the priority of the vertex cache.

Finally, all VCNs upload their weights to gNB, and gNB selects appropriate VCNs as the cache nodes of the content according to the weights of all VCNs. The flow of the algorithm is shown in Algorithm 2. In the worst case, the time complexity of the algorithm is $O(N^f \times N^v \times N^t)$, where N^f is the number of contents, N^v is the number of VCNs, and N^t is the number of time slots.

V. Performance Evaluation

A. Simulation Setup

In this section, we evaluate the performance of the proposed caching strategy by numerical simulation in Python. It is worth mentioning that our proposed ACC strategy uses two different adaptive caching strategies, namely, centralized caching decisions for the RCN layer and distributed caching decisions for the VCN layer. We designed and simulated a real simulation environment with the simulation parameters shown in Table II. In the designed scenario, the length of the road is 2 km and the width is 15 m. The number of RCNs placed along the road is 4, the coverage area is 500 m, and the cache capacity is 4GByte. The number of VCNs on the road is 30, the communication range of VCNs is 100 m, the cache capacity is 1GByte, and the velocity ranges from 8 to 12 m/s. The size of each content in the gNB is set to 50 to 500 MB, and the transmission rates are r^{vv}=12 Mbps, r^{vi}=10 Mbps, and r^{vn}=6 Mbps, respectively. And the transmission delay of downloaded content can be calculated by the above (9). We assume that the probability of each vehicle user requesting content obeys Zipf's law distribution [46].

TABLE II
Simulation Parameters

Parameter	*Value*
Number of RCNs N^r	4
Number of VCNs N^v	30
Number of contents N^f	50-500
Cache capacity of the RCN S^r	2-9 GByte
Cache capacity of the VCN S^v	0.5-4 GByte
Size of content S^f	50-500 MByte
Velocity of vehicle u_i	8-12 m/s
Size of the packet S^p	1 MByte
Transmission rate of V2V r^{vv}	12 Mbps
Transmission rate of V2I r^{vi}	10 Mbps
Transmission rate of V2N r^{vn}	6 Mbps
Zipf parameters γ	0.1-0.8
Communication range of RCNs	500 m
Communication range of VCNs	100 m
Length of road	2 km
Width of road	15 m

To verify the performance of the ACC strategy, we considered the following five caching strategies as comparison experiments:

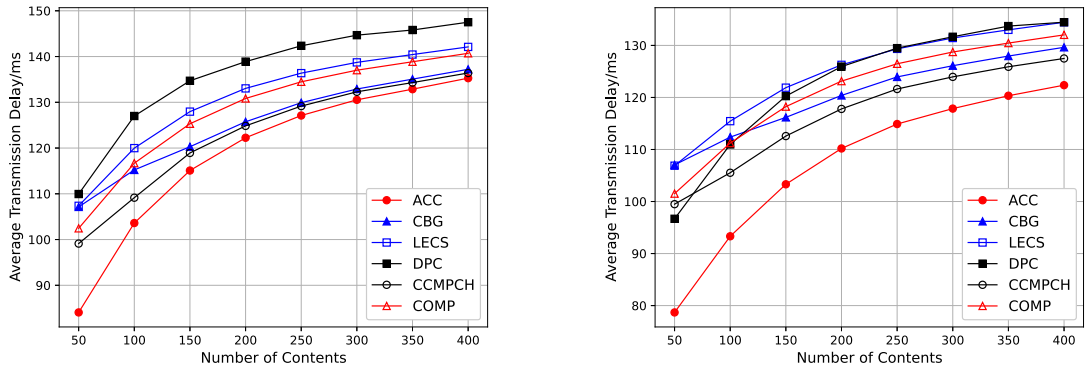
1) DPC [30]: DPC adopts a distributed caching mode, all RCNs and VCNs do not cooperate and perform caching decisions only based on content popularity and local caching status.
2) LECS [29]: The LECS strategy only considers the caching strategy of the RCN layer and evaluates the value of all contents, with the goal of maximizing the value of the system.
3) COMP [11]: COMP divides all VCNs into appropriate clusters, and the cluster head of each cluster is responsible for caching and distributing data.
4) CBG [8]: CBG considers cooperative caching between RCNs and VCNs, and all the contents cached by RCNs and VCNs are not repeated.
5) CCMPCH [39]: CCMPCH considers cooperative caching between RCNs and VCNs, and divides VCNs into different clusters based on the properties of vehicles, and the cached content within each cluster is not repeated.

Next, we evaluate the above strategy in terms of three performance metrics: average transmission delay, cache hit ratio and communication overhead for the whole road system, respectively.

B. Average Transmission Delay

Transmission delay is one of the important indicators to measure the performance of communication systems. Transmission delay can affect the reliability and efficiency of communication between vehicles. For example, in a safety-critical situation, where a vehicle needs to send an emergency message to other vehicles, a delay in transmission could cause a serious accident. In addition, transmission delay can also affect the quality of service (QoS), different applications have different requirements

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(a) $\gamma = 0.2$ (b) $\gamma = 0.6$

Fig. 4. Average transmission delay versus number of contents.

for QoS. For example, safety applications require low delay and high reliability, while entertainment applications may have more relaxed QoS requirements.

Therefore, in this section, we will discuss the average transmission delay of vehicle user access content under different caching policies. As shown in Fig. 4, we discuss the average transmission delay of vehicle users accessing content under different caching policies with different Zipf parameters as the number of contents changes in gNB when the cache capacity of RCN and VCN remains unchanged (S^r=4GByte, S^v=1GByte). In a diverse environment, such as UCB, QuestNet, and NLANR, the popularity of content approximately follows a Zipf parameter γ centered around the parameter 0.7 [46], [8], [39]. Furthermore, to assess the scalability of caching strategies, we also need to consider the performance metrics of different caching strategies in the poor case (i.e., when γ are lower). Considering that vehicular edge networks are highly dynamic, we set the Zipf parameter γ to 0.6 (general) and 0.2 (poor) in our simulation experiments. From Fig. 4, it can be seen that compared with the other four caching policies, our proposed ACC policy can significantly reduce the average transmission delay of vehicle users. Additionally, it can be observed that as the number of contents in gNB increases, the average transmission delay of all caching policies will increase. This is because as the number of contents in gNB increases, the probability of vehicle users accessing content cached in VCN or RCN decreases, making it more likely for vehicle users to obtain content from gNB, thereby increasing the average transmission delay.

Then we discussed the changes in average transmission delay with respect to the Zipf parameter γ under different caching strategies. Fig. 5 shows the changes in average transmission delay concerning the Zipf parameter γ under different caching strategies when the number of contents N^f in gNB is 200, S^r=4GByte and S^v=1GByte. From Fig. 5, it can be seen that our proposed ACC strategy is significantly better than other caching strategies regardless of the value of γ. In addition, we can also observe that as the Zipf parameter γ increases, the average transmission delay of all caching strategies decreases. This is because as γ increases, the popularity of content becomes more concentrated on a few contents. When caching the same number of contents, a larger γ leads to an increase in the sum

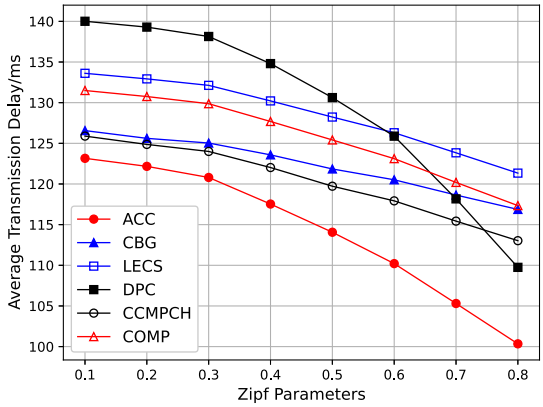


Fig. 5. Average transmission delay versus Zipf parameters.

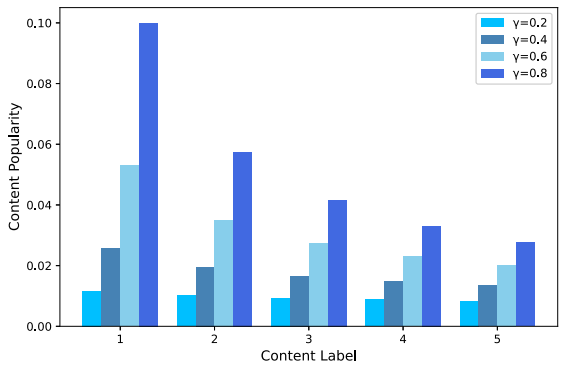
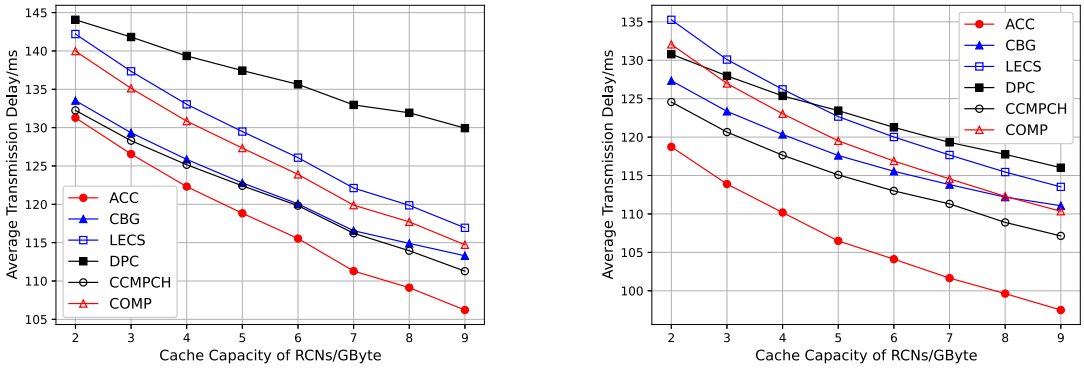


Fig. 6. Content popularity versus Zipf parameters.

of the popularity of content in the cache, making it more likely for vehicle users to retrieve content from the cache and thus reducing overall average transmission delay.

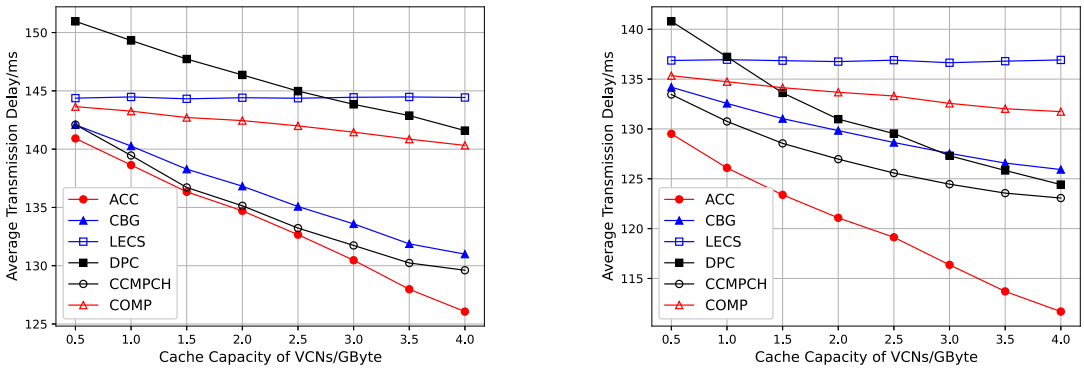
Furthermore, The change in content popularity with Zipf parameter γ is shown in Fig. 6, which displays the change in the popularity of the top five ranked contents under different Zipf parameters (γ from 0.2 to 0.8) when the total number of contents in gNB is 200. From Fig. 6, it can be seen that the sum of the popularity of the top five ranked contents increases as γ increases. Therefore, when caching the same number of

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(a) $\gamma = 0.2$ (b) $\gamma = 0.6$

Fig. 7. Average transmission delay versus cache capacity of RCNs.



(a) $\gamma = 0.2$ (b) $\gamma = 0.6$

Fig. 8. Average transmission delay versus cache capacity of VCNs.

contents, a larger γ results in a higher sum of content popularity in the cache, making it more likely for vehicle users to access content from the cache, thereby reducing the overall average transmission delay.

In addition, we also discuss the impact of the cache capacity of RCN S^r and VCN S^v on the overall average transmission delay, as shown in Figs. 7 and 8. Fig. 7 shows the overall average transmission delay under different caching policies with RCN cache capacity S^r ranging from 2-9GByte when the total number of contents in gNB is 200 and the Zipf parameter γ is 0.2 and 0.6, respectively. Firstly, it can be observed that compared with the other four caching policies, our proposed ACC policy can significantly reduce the overall average transmission delay of vehicle users regardless of the value of γ. From Fig. 7, it can be seen that as the S^r increases, the overall average transmission delay of all caching policies will decrease. This is because as the S^r increases, the number of contents cached in RCN will also increase, making it more likely for vehicle users to obtain content from the RCN cache, thereby reducing the overall average transmission delay.

Fig. 8 shows the overall average transmission delay under different caching strategies when the total number of contents in gNB is 500 and the Zipf parameter γ is 0.2 and 0.8, and the cache capacity S^v of VCN varies from 0.5-4GByte. First, it can be observed that ACC is consistently better than other caching strategies. From Fig. 8, it can be seen that as S^v increases, the overall average transmission delay of all caching strategies decreases. This is also because as S^v increases, the number of contents cached in VCN also increases, making it more likely for vehicle users to retrieve content from the VCN cache and thus reducing the overall average transmission delay. It is worth mentioning that the LECS strategy remains the same because he only considers the RCN layer cache.

C. Cache Hit Ratio

In edge cooperative caching, frequently accessed data is cached at the vehicle or the RSU, allowing for faster retrieval times and reducing the number of requests that need to be sent over the network. A high cache hit ratio indicates that a large percentage of requests are being served from the cache, which can lead to reduced network traffic and improved system performance. A lower cache hit ratio suggests that the cache is not effectively storing frequently accessed data, leading to increased delay and more frequent access to the original data source. Therefore, the cache hit ratio is a key performance metric for evaluating the effectiveness of edge cooperative caching. It

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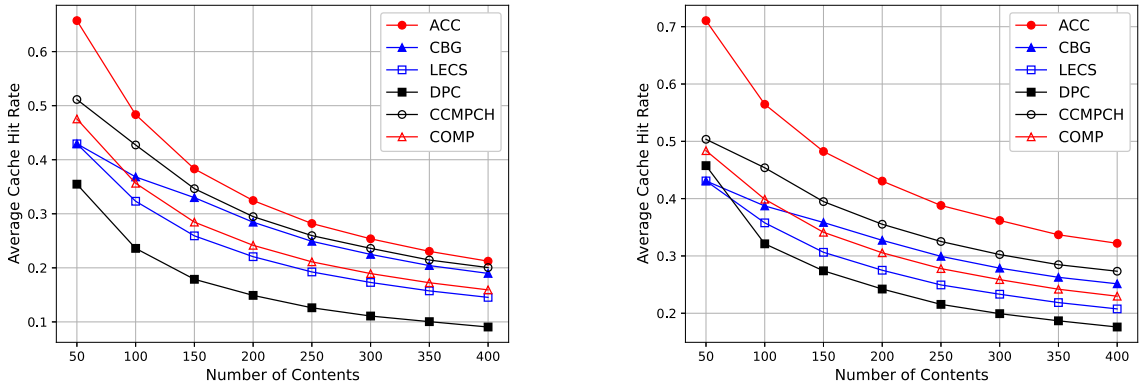


Fig. 9. Average cache hit ratio versus number of content.

measures the percentage of requests that are served from the cache, rather than being sent to original data source.

The cache hit ratio is calculated by dividing the number of cache hits by the total number of requests. In this paper, the cache hit ratio can be divided into the cache hit ratio of the VCN layer H^v and the cache hit ratio of the RCN layer H^r which can be calculated separately as:

$$H^v = \frac{N_v^h}{N^h}, \tag{25}$$

$$H^r = \frac{N_r^h}{N^h}, \tag{26}$$

where N_v^h and N_r^h are the numbers of cache hits to the VCN layer and the RCN layer, respectively, and N^h is the total number of cache hits. Thus the average cache hit ratio H_{avg} can be calculated as:

$$H_{avg} = \frac{\alpha_1 N_v^h + \alpha_2 N_r^h}{N^h}, \tag{27}$$

where α_1 and α_2 are the weights of the VCN layer and the RCN layer, respectively, and $\alpha_1 + \alpha_2 = 1$.

Therefore, we first discuss the relationship between the average cache hit ratio and the number of contents in Fig. 9, it shows the change in the average cache hit ratio of all strategies with the variation of the total content in gNB, when the cache capacity of RCN and VCN remains constant and Zipf parameters are 0.2 and 0.6, respectively. As can be seen from the figure, with an increase in the total number of contents, the cache hit ratio of all caching strategies decreases. This is because, with a constant cache capacity, the proportion of content cached at edge nodes decreases with an increase in the total number of contents, leading to a decrease in the cache hit ratio. It is worth mentioning that regardless of the total number of contents, the cache hit ratio of the proposed ACC strategy is always higher than all other strategies.

Then, we discuss the relationship between the average cache hit ratio and the Zipf parameters in Fig. 10, it shows the change in the average cache hit ratio of all strategies with the variation of the Zipf parameter γ from 0.1 to 0.8, when the total number of contents in gNB is 200 and the cache capacity of RCN and VCN remains constant. As can be seen from Fig. 10, with an increase in γ, the cache hit ratio of all caching strategies increases. This is because as γ increases, the popularity of the top-ranked contents increases, leading to an increase in cache hit ratio. It is worth mentioning that regardless of the value of γ, the cache hit ratio of the proposed ACC strategy is always higher than all other strategies, and the improvement in hit ratio becomes more significant with an increase in γ.

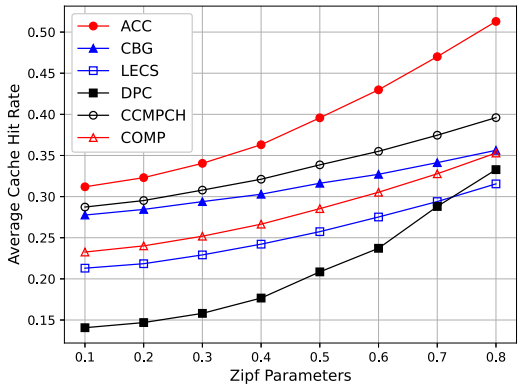
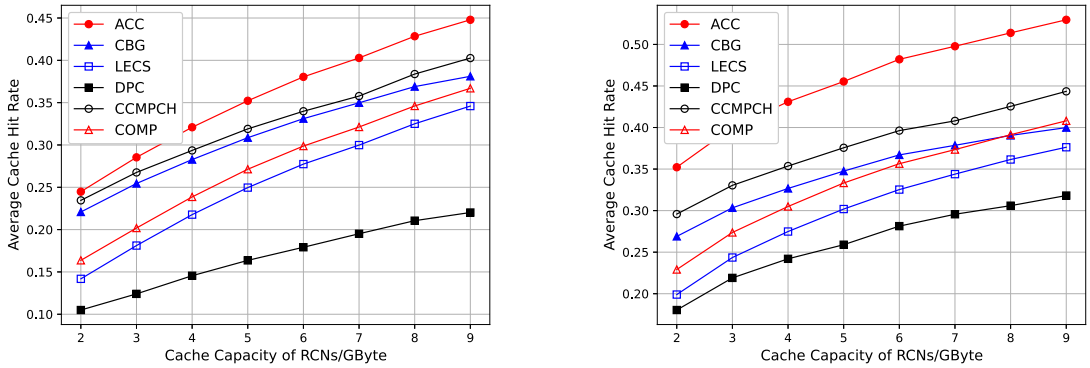


Fig. 10. Content popularity versus Zipf parameters.

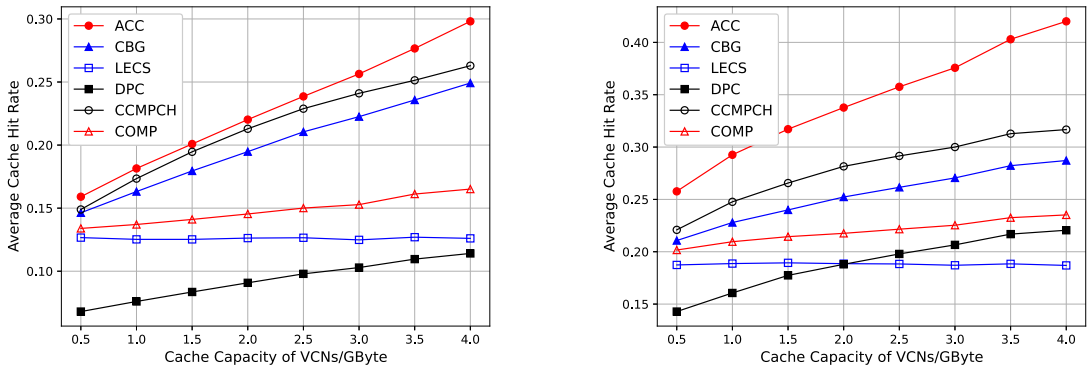
Finally, we discuss the relationship between the average cache hit ratio and the cache capacity of RCN and VCN. Fig. 11 shows the change in the average cache hit ratio of all strategies with the variation of the cache capacity of RCN S^r, when the total number of contents is 200 and the Zipf parameters are 0.2 and 0.6. Fig. 12 shows the change in the average cache hit ratio of all strategies with the variation of the cache capacity of VCN S^v, when the total number of contents is 500 and the Zipf parameters are 0.2 and 0.6. As can be seen from Figs. 11 and 12, with an increase in the S^r or S^v, the cache hit ratio of all caching strategies increases. This is because as the S^r or S^v increases, the number of contents cached in RCN or VCN also increases, making it more likely for vehicle users to retrieve content from RCN or VCN cache and thus increasing the cache hit ratio. The cache hit ratio of the ACC strategy is always higher than that of the other strategies, especially when the Zipf parameter is large (γ=0.6).

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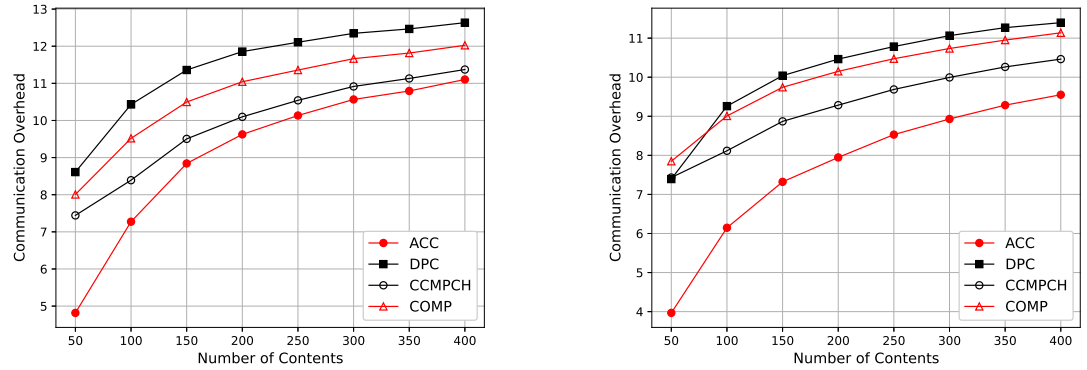
(a) $\gamma = 0.2$ (b) $\gamma = 0.6$

Fig. 11. Average cache hit ratio versus cache capacity of RCNs.



(a) $\gamma = 0.2$ (b) $\gamma = 0.6$

Fig. 12. Average cache hit ratio versus cache capacity of VCNs.



(a) $\gamma = 0.2$ (b) $\gamma = 0.6$

Fig. 13. Communication overhead versus number of content.

D. Communication Overhead

Finally, we also evaluate the communication overhead under different caching strategies. We define the communication overhead under VNDN architecture as the ratio of the number of interest packets forwarded in the network to the number of data packets [39]. Therefore, we discuss the relationship between communication overhead and the number of contents in Fig. 13. Fig. 13 shows how the communication overhead varies with the total amount of content in gNB under the VNDN based caching strategy when the cache capacity remains constant and the Zipf parameter γ is 0.2 and 0.6, respectively. In the figure, we can see that the communication cost increases with the increase of the total number of contents. This is because when the cache

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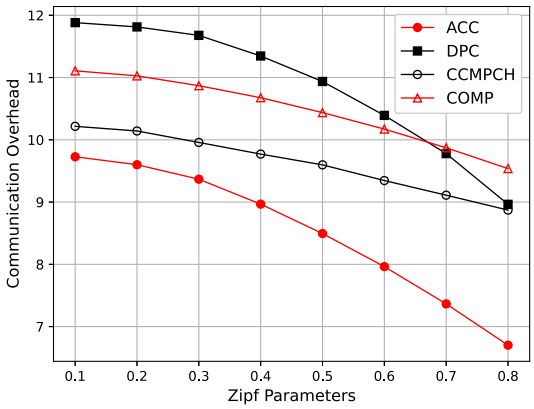


Fig. 14. Communication overhead versus Zipf parameters.

capacity is unchanged, with the increase of the total number of contents, more contents need to be forwarded by the intermediate node several times before they can be obtained by the vehicle user, resulting in an increase in the communication cost. It is not difficult to see from the figure that the communication overhead of our proposed ACC strategy is always lower than that of other VNDN based caching strategies.

In addition, we discuss the relationship between the communication overhead and the Zipf parameter γ in Fig. 14, which shows that the communication overhead decreases with the increase of the Zipf parameter while other parameters are held constant. This is because the larger γ, the more concentrated the distribution of content popularity, and the highly popular content has a higher cache priority, so vehicle users often only need to forward a small number of interest packages to obtain the highly popular content, thus reducing the overall communication overhead. As can be seen from the figure, the communication overhead of our proposed ACC strategy is also always lower than that of other VNDN based caching strategies.

VI. Conclusion

In this paper, we propose an adaptive cooperative caching strategy for vehicular networks, which divides edge nodes into a fixed RCN layer and a mobile VCN layer, using centralized and distributed caching strategies, respectively, and can be adaptively adjusted based on content popularity and vehicle mobility. We minimize the average transmission delay as the overall optimization objective and divide it into two sub-tasks, namely, the minimization of average transmission delay in the RCN layer and the minimization of average transmission delay in the VCN layer. Then, we transform the optimization problem of the RCN layer into an MCK problem and the optimization problem of the VCN layer into an MMWDS problem, respectively, and use heuristic algorithms with low complexity to obtain approximate solutions for optimal caching decisions. Simulation results show that the proposed strategy can effectively improve cache hit ratio and reduce average transmission delay compared to other caching strategies. However, the performance of the proposed strategy largely depends on the prediction of content popularity and vehicle mobility. Therefore, in future work, we will consider how to better predict content popularity and vehicle mobility to further improve the performance of the proposed strategy.

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